GPU-Accelerated Numerical Analysis of Blood Flow Through Arteries Using the Finite Volume Method (FVM)

Objective and Scope:

The primary objective of this report is to provide a thorough analysis of blood flow through arteries utilizing the Finite Volume Method. This investigation seeks to deepen our understanding of how various geometric and physiological factors impact crucial hemodynamic parameters. The report specifically focuses on key areas, including:

The sensitivity of coronary hemodynamic to the structural characteristics of blood vessels encompasses factors such as bifurcation angles, positional relationships, diameter ratios, and the tortuosity of arteries. The influence of stenotic lesions—narrowing in the arterial path—on flow patterns, pressure distributions, and wall shear stress is vital for understanding cardiovascular health. Implementing non-Newtonian fluid models is essential for accurately representing the complex rheological behaviour of blood, which is essential for realistic simulations.

The application of non-Newtonian fluid models to capture the complex rheological behaviour of blood. This report contributes to the growing body of knowledge on arterial hemodynamic and its implications for cardiovascular health and disease by synthesizing theoretical frameworks with computational simulations and experimental validations. The findings have potential applications in predicting disease progression, designing medical interventions, and developing personalized treatment strategies based on patient-specific vascular geometries.

Left Anterior Descending Artery (LAD) Specifications

The Left Anterior Descending (LAD) artery is the principal branch of the left coronary artery (LCA), responsible for supplying oxygenated blood to the anterior myocardium. Its clinical significance is underscored by its designation as the "widow-maker" artery due to the high mortality risk associated with proximal occlusions.

Segmental Classification

For precise hemodynamic analysis, the LAD is divided into four anatomical segments:

1. **Proximal LAD** (origin to first diagonal branch D1)
2. **Mid-Proximal LAD** (D1 to D2)
3. **Mid-Distal LAD** (D2 to D3)
4. **Distal LAD** (beyond D3 to cardiac apex)

Geometric and Hemodynamic Parameters

This study focuses on the proximal LAD segment with the following characteristics:

- **Density of blood**: 1060 kg/m^3
- **Diameter:** 3.5 mm (0.0035 m) [Range: 3.1-3.7 mm based on clinical data]
- **Length:** 14 cm (0.14 m) [Anatomically representative range: 10-15 cm]
- **Time-Averaged Mean Velocity:** 0.39 m/s (39 cm/s) [Physiological range: 0.15-0.40 m/s]

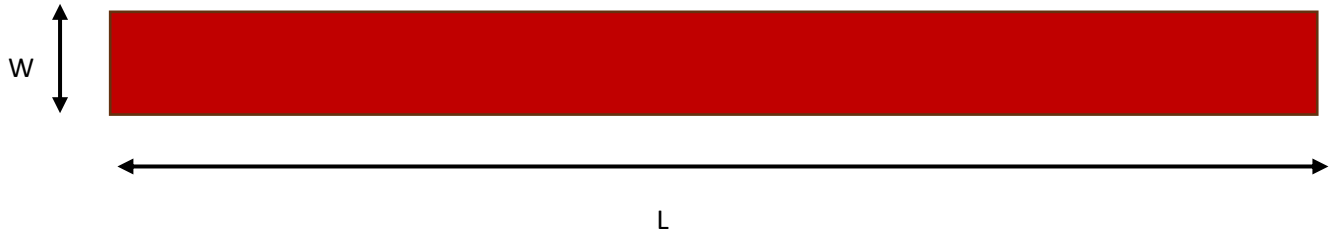
***Data reference**:* Ali, Sajid, et al. [2024], *[Journal of X-Ray Science and Technology]*

Numerical Study:

Numerical Study Conditions

1. **Flow Assumptions**
    - The flow is modelled as unsteady, incompressible, and laminar, as blood flow is a time-dependent physiological process.
    - The non-Newtonian, shear-thinning behaviour of blood is considered, where viscosity decreases with increasing shear rate.
2. **Computational Domain**
    - A 2D Cartesian grid is employed for the numerical simulations.
    - The flow direction is defined from left to right (positive *x*-direction, $Ue>0$).
3. **Rheological Model**
    - The Bird-Carreau model is used to characterize the shear-thinning behaviour of blood, with viscosity defined
4. **Boundary Conditions**
    - **Inlet:** inlet velocity profile as constant inflow.
    - **Outlet:** Pressure-based boundary condition (e.g., zero pressure gradient).
    - **Walls:** No-slip condition ($u=0u=0$).
5. **Numerical Scheme**
    - The Finite Volume Method (FVM) is applied for spatial discretization.
    - A first-order Euler explicit method is used to handle the unsteady flow characteristics.

Geometry of the artery:



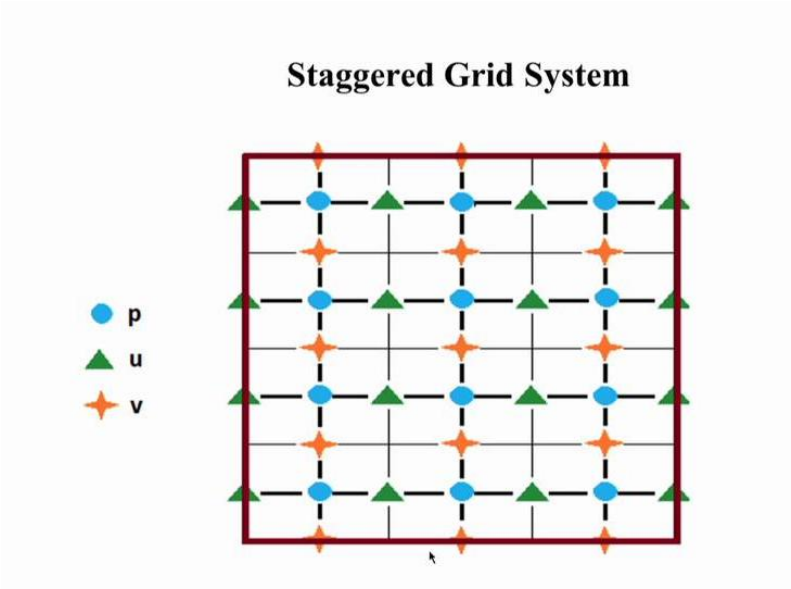
Domain Dimensions:

- Length = 0.14 m
- Width = 0.0035m
- Cell size in x directions = 0.0001 m
- Cell size in y direction = 0.0001 m

Staggered Grid Approach:

The **staggered grid** is used for spatial discretization method

- **Velocity components** (u, v) are stored at **cell faces**.
- **Pressure (p) and scalar quantities** (e.g., viscosity) are stored at cell center.



Spatial Discretization & Iterative Solution Methods

1. Convection Term: QUICK Scheme

- Scheme: Third-order QUICK (Quadratic Upstream Interpolation for Convective Kinematics)
- Formulation:
  For a generic variable ϕ (e.g., velocity component):

$$\phi f = \{\phi UP + 8\psi(3\phi D - 2\phi UP - \phi UU)$$

2. Diffusion Term: Central Difference Scheme

- Scheme: Second-order Central Difference

3.Validation & Stability:

- **Grid Sensitivity**: Verify QUICK accuracy via grid refinement studies.
- **CFL Condition**: For explicit convection, CFL=uΔt/Δx≤ 1
- **Shear-Thinning**: Ensure η(γ˙)$\eta(\gamma\dot{})$ is updated iteratively within GS loops.